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AI-POWERED INFRASTRUCTURE INTELLIGENCE FOR ZERO-FAILURE MINING & CIVIL OPERATIONS

PREDICTIVE MONITORING • REMOTE-READY • COMPLIANCE-DRIVEN • MISSION-CRITICAL

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OUR VISION & MISSION

OUR VISION

TO BECOME THE WORLD'S MOST TRUSTED AI-DRIVEN INFRASTRUCTURE INTELLIGENCE PLATFORM, ENSURING ZERO CATASTROPHIC FAILURES ACROSS MINING AND CIVIL ENGINEERING ENVIRONMENTS.

OUR MISSION

To empower infrastructure operators with:

- Real-time structural awareness
- Predictive early-warning intelligence
- Automated compliance & audit systems
- AI-driven decision support

Making infrastructure safer, more resilient, and operationally efficient — even in remote and harsh environments.

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WHO WE ARE

OCTASENCE IS A NEXT-GENERATION AGENTIC AI-POWERED GEOTECHNICAL INTELLIGENCE & STRUCTURAL HEALTH MONITORING (SHM) PLATFORM BUILT FOR HIGH-RISK INFRASTRUCTURE.

WE COMBINE

- AI Agents
- Multimodal IoT Sensors
- Geotechnical Modeling
- Edge Computing
- Real-Time Digital Twins

WE SERVE

- Underground & open-pit mining
- Tailings & dams
- Tunnels & metro systems
- Rail & transport infrastructure
- Large civil & public assets

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THE CUSTOMER PROBLEM

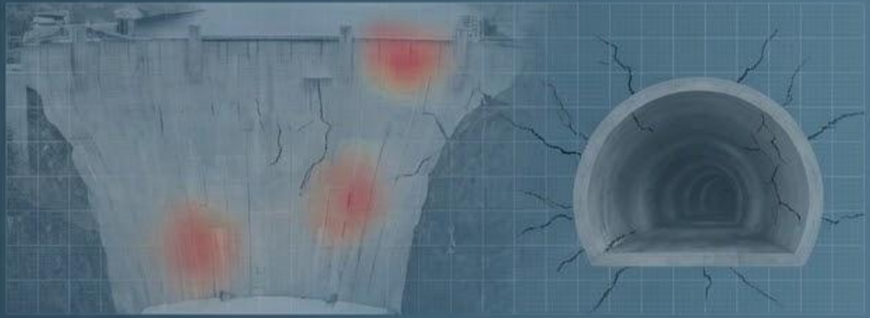
Infrastructure failures are:

SUDDEN
LIFE-THREATENING
ENVIRONMENTALLY DAMAGING
FINANCIALLY CATASTROPHIC
OFTEN PREVENTABLE

CURRENT REALITY

Most organizations still rely on:

- Periodic inspections
- Threshold-based alarms
- Fragmented sensor systems
- Manual expert interpretation
- Cloud-only monitoring (fails in remote areas)



THE REAL FAILURE:

Not structural — informational.

Risk becomes visible only when failure has already begun.

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WHERE WE WORK

MINING
- Pillar & roof collapse prediction
- Bench & slope stability monitoring
- Haul road & embankment stability

DAMS & TAILINGS
- Seepage & pore pressure monitoring
- Liquefaction risk
- Breach early-warning systems

TUNNELS & UNDERGROUND
- Settlement detection
- Water ingress risk
- Structural strain monitoring

RAIL & TRANSPORT
- Track misalignment
- Embankment washouts
- Bridge fatigue

REMOTE & LOW CONNECTIVITY SITES
- Desert mines (WA, Africa)
- Mountain tunnels
- Offshore & isolated facilities

Built to operate reliably even with satellite-only connectivity.

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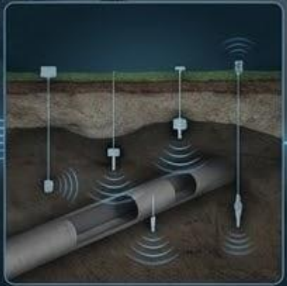
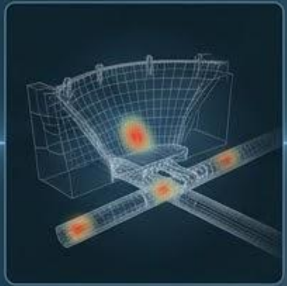
HOW OUR SERVICE WORKS





STEP 1 — MULTIMODAL SENSOR DEPLOYMENT

Sensors measure:

- Strain
- Stress
- Displacement
- Tilt
- Vibration
- Seepage
- Groundwater
- Environmental conditions

STEP 2 — EDGE INTELLIGENCE

- On-site AI inference
- Data buffering
- Noise filtering
- Offline-first architecture
- Store-and-forward transmission

STEP 3 — AI & GEOTECHNICAL MODELING

- Time-series anomaly detection
- Failure probability prediction (6–72 hours window)
- Physics-informed modeling
- Explainable risk scoring

STEP 4 — DIGITAL TWIN VISUALIZATION

- Live 2D/3D asset models
- Risk heatmaps
- Failure propagation simulation

STEP 5 — ALERTS & PREVENTIVE ACTION

- WhatsApp / SMS / Email / Dashboard
- Severity-based escalation
- AI-recommended corrective action

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CORE SERVICES

1 REAL-TIME MONITORING

- Continuous structural & geotechnical sensing
- Live risk dashboards
- Multi-asset visibility

2 PREDICTIVE FAILURE ANALYTICS

- Pillar collapse prediction
- Slope instability forecasting
- Tailings breach risk analysis
- Tunnel deformation alerts

3 EDGE + REMOTE MONITORING

- Offline-first architecture
- Satellite-compatible transmission
- Remote-region resilience

4 INFRASTRUCTURE DIGITAL TWINS

- Live structural models
- Simulation-based decision support
- Time-lapse risk evolution

5 COMPLIANCE & AUDIT AUTOMATION

- DGMS / ISO / BIS aligned reports
- Full audit trail
- Daily AI-generated safety summaries

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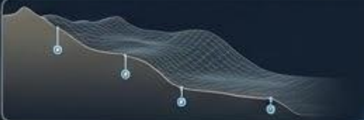
REMOTE MONITORING ARCHITECTURE
(Ultra-Remote Mining & Satellite Connectivity Model)

REMOTE MINING REALITY

Maoy mining sites operate in:
- Western Australie desert regions
- African remote belts
- Mountainous tunnols
- Offshore & isolated civil projects

Challenges:
- No terrestrial fiber
- Ne celiular coverege
- Extreme temperatare & dust
- Limited grid power
- Satellite only connectivity

SITE SENSOR LAYER



Geotechnical & strustural sensors deployed across:
- Slopes
- Pillars
- Tailings
- Tailings
- Tunnels
- Embankments

EDGE GATEWAY LAYER

Industrial edge device performs:
- Local data buffering (30–60 days)
- Noise fifiering & validafien
- Low-latency anomaly detection
- Soler + battery supported

SATELLITE COMMUNICATION

Data trensmitted via:
- Starluik
- SkyMesh
- Pivotal
- Nexion Group

Sopporte:
- Full date sync (normal mode)
- Alert only low bandwidth mode
- Encrypted secure transmission

OCTASENCE CLOUD SAAS

- AI processing
- Risk scoring
- Digital twin visuallyation
- Compliance reporting

REMOTE ACCESS

HQ / Control Room Access from:
- Perth
- Sydney
- CU oparations centers
- Government dashbocards

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SENSOR & TECHNOLOGY PARTNERS

OctaSence collaborates with globally recognized instrumentation providers.

GEOTECHNICAL & STRUCTURAL SENSORS

HBK
- Strain gauges
- Structural load monitoring
- Fatigue measurement

GroundProbe®
- Slope stability radar
- Open-pit deformation monitoring

Encardio Rite
- Piezometers
- Inclinometers
- Settlement systems

BWSENSING
- Distributed fiber optic sensing
- Strain & temperature profiling

SATELLITE & REMOTE NETWORKING PARTNERS

STARLINK

SkyMesh

Pivotel

NEXION GROUP

Sensor-agnostic. Network-agnostic. Integration-ready.

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TECHNICAL POSITIONING

(Site → Sensors → Edge → SaaS Intelligence Architecture)

1 SITE & PHYSICAL ASSET LAYER

Assets monitored:

- Open-pit benches
- Underground pillars
- Tailings dams
- Tunnels
- Embankments
- Bridges

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2 SENSOR & DATA ACQUISITION LAYER

Multimodal instrumentation:

- Strain & stress
- Tilt & displacement
- Vibration
- Pore pressure
- Seepage
- Environmental conditions
- Vision-based structural analysis

Continuous sampling + time-synchronized logging.

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3 EDGE INTELLIGENCE LAYER

Industrial-grade gateways:

- Embedded AI inference
- Anomaly detection at source
- Data compression
- Fail-safe storage
- Secure encryption
- Solar-compatible deployment

Offline-first capability ensures: No data loss during connectivity interruptions.

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4 CLOUD AI & SAAS INTELLIGENCE LAYER

Core Capabilities:

- Failure probability modeling
- Time-series forecasting
- Rate-of-change analysis
- Stress–strain modeling
- Rainfall correlation logic
- Risk scoring engine

Outputs:

- Digital twin heatmaps
- Escalation alerts
- Automated compliance reports
- Audit traceability logs

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WHY CHOOSE OCTASENCE

✓ **BUILT SPECIFICALLY FOR HIGH-RISK ENVIRONMENTS**
(Designed for mining, dams, tunnels, and critical infrastructure)

✓ **PREDICTIVE — NOT REACTIVE**
(6–72 hour early-warning capability)

✓ **OPERATES IN REMOTE & SATELLITE-ONLY SITES**
(Unreliable connectivity is no barrier to monitoring)

✓ **SENSOR-AGNOSTIC INTEGRATION**
(Compatible with existing or new sensor instrumentation)

✓ **COMBINES AI + GEOTECHNICAL DOMAIN LOGIC**
(Reduces false alarms and improves model accuracy)

✓ **ENTERPRISE & REGULATOR READY**
(Aligned with global industrial and safety standards)

✓ **SCALES FROM SINGLE ASSET TO MULTI-SITE DEPLOYMENT**
(Growth-ready architecture for large operations)

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WHAT MAKES US DIFFERENT

BUILT FOR PREDICTION — NOT MONITORING

Failure forecasting, not threshold alarms.

UNIFIED INTELLIGENCE STACK

AI Agents + IoT + Geotechnical Models + Digital Twins.

REMOTE-READY ARCHITECTURE

Edge inference + satellite connectivity support.

EXPLAINABLE AI

Transparent risk reasoning — not black-box outputs.

COMPLIANCE BY DESIGN

Built for regulators, insurers, and board-level reporting.

HIGH DEFENSIBILITY

Hardware + software + domain expertise combined.

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CORE CUSTOMER BENEFITS

PREVENT CATASTROPHIC FAILURES

Early warning before collapse or breach.

REDUCE OPERATIONAL DOWNTIME

Avoid shutdowns & regulatory penalties.

IMPROVE WORKER SAFETY

Real-time alerts reduce human exposure.

LOWER INSURANCE & COMPLIANCE RISK

Prove continuous due diligence.

OPERATE IN REMOTE ENVIRONMENTS

Maintain monitoring even without terrestrial network.

BETTER EXECUTIVE VISIBILITY

Enterprise dashboards across multiple sites.

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SCALABLE DEPLOYMENT MODEL

PILOT ASSET
Single slope, TSF, tunnel, or bridge section validation.

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MULTI-ASSET WITHIN SITE
Full site-level structural intelligence.

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MULTI-SITE WITHIN ENTERPRISE
Enterprise-wide dashboard integration.

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NATIONAL / GOVERNMENT-LEVEL DEPLOYMENT
Regulatory-grade infrastructure intelligence network.

HIGH RETENTION DUE TO:

- Data continuity
- Compliance dependency
- Operational workflow integration

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PILOT PARTNERSHIP FRAMEWORK

PHASE 1 — PILOT PARTNERSHIP ALIGNMENT

Key Activities:
- Asset selection (e.g., slope, TSF, tunnel section)
- Risk objective definition
- Stakeholder alignment
- Pilot duration agreement (8–12 weeks)

Deliverable:
Pilot scope document + success metrics.

→

PHASE 2 — SITE / STRUCTURE TECHNICAL EVALUATION

Key Activities:
- Geotechnical assessment
- Asset geometry mapping
- Environmental risk profiling
- Existing sensor audit
- Connectivity analysis

Deliverable:
Technical requirement report.

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PHASE 3 — REQUIREMENT ANALYSIS & SYSTEM DESIGN

Key Activities:
- Sensor type finalization
- Sampling rate planning
- Edge device sizing
- Satellite bandwidth planning
- Power autonomy calculation

Deliverable:
Deployment architecture blueprint.

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PHASE 4 — SENSOR INSTALLATION & CALIBRATION

Key Activities:
- Instrument placement
- Baseline data capture
- Calibration & validation
- Health-check configuration

Deliverable:
Commissioning report.

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PHASE 5 — NETWORKING & DATA INGESTION

Key Activities:
- Satellite integration
- Edge configuration
- Data encryption setup
- SaaS platform onboarding
- Alert channel configuration

Deliverable:
Live monitoring dashboard activation.

→

PHASE 6 — SAAS MONITORING & INTELLIGENCE VALIDATION

Key Activities:
- AI model calibration
- Risk scoring activation
- Alert testing & verification
- Weekly engineering review
- Performance reporting

Deliverable:
Pilot evaluation report + rollout roadmap.

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ENGAGEMENT MODEL

AI

PHASE 1 — **ASSESSMENT**

- Site evaluation
- Asset prioritization
- Sensor mapping
- Risk baseline

PHASE 2 — **PILOT DEPLOYMENT**

- 1 selected asset
- 8–12 week monitoring cycle
- Data validation & model calibration

PHASE 3 — **FULL DEPLOYMENT**

- Multi-asset rollout
- Enterprise dashboard setup
- Integration with SCADA / ERP

PHASE 4 — **LONG-TERM MONITORING**

- Annual SaaS subscription
- Ongoing AI model refinement
- Continuous compliance reporting

COMMERCIAL MODEL

- SaaS subscription (per asset / per site)
- Hardware (sale or lease)
- Deployment & integration services
- Enterprise contracts

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OUR COMMITMENT

- Zero catastrophic failures
- Engineering-grade reliability
- Transparent AI reasoning
- Regulatory alignment
- Continuous innovation
- Long-term partnership

We build solutions for real mines, real dams, real tunnels — not lab simulations.

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LET'S CONNECT

Contact OctaSence

AI-Powered Infrastructure Intelligence For Safer, Smarter, Resilient Operations

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